EcoSort AI: CNN-Based Waste Classification for Sustainability

The growing challenge of waste management has become one of the key issues affecting environmental sustainability. With rapid urbanization and industrial development, waste generation continues to rise, causing land, water, and air pollution. Traditional waste segregation methods are inefficient, labor-intensive, and prone to errors. To overcome these challenges, modern technological approaches like Artificial Intelligence (AI) and Deep Learning can be leveraged to create automated and accurate waste classification systems.

Project Problem Statement

This project, titled **EcoSort AI**, focuses on developing an intelligent waste classification model using **Convolutional Neural Networks (CNNs)**. The system aims to automatically identify and categorize waste images into six classes: cardboard, glass, metal, paper, plastic, and trash. By integrating CNN-based deep learning techniques, the project promotes automation in waste segregation processes and enhances recycling efficiency. The model will be trained using the **Garbage Classification Dataset** from Kaggle and refined using **Google Teachable Machine** for improved accuracy and adaptability.

The primary objective of EcoSort AI is to contribute to sustainable development by utilizing deep learning for smarter waste management. The project aligns with the **United Nations Sustainable Development Goal 12 – Responsible Consumption and Production**, encouraging eco-friendly practices and reducing landfill waste through intelligent automation.

Dataset

Dataset Name: Garbage Classification Dataset
Source: Kaggle – A high-quality image dataset designed for waste classification and recycling research.

This dataset contains thousands of labeled images categorized into six waste types: cardboard, glass, metal, paper, plastic, and trash. Each image is pre-organized into folders, making it suitable for CNN-based classification tasks. It provides an ideal foundation for developing AI-powered waste segregation systems that support sustainable environmental practices.

Next Steps

- 1. Collect and Prepare Dataset from Kaggle.
- 2. Preprocess and organize images for training and validation.
- 3. Train the CNN Model using Google Teachable Machine.

- 4. Evaluate the Model’s Performance using accuracy metrics and confusion matrix.
- 5. Build a Simple Web Interface using Python Streamlit for testing classifications.
- 6. Test the Application with real-world images of waste materials.
- 7. Deploy and Document the Complete Project.

EcoSort AI represents an innovative approach to merging technology with environmental responsibility. By applying CNN and deep learning methods to waste classification, this project demonstrates how AI can be a powerful tool for achieving a cleaner, greener, and more sustainable future.